

Ultimate Math Formula Sheet

Everything the Digital SAT gives you on the reference sheet — plus the formulas it makes you memorize.

Given on the SAT Reference Sheet (Geometry)

These appear inside Bluebook on every math question. You still need to know when to use them.

Shape / Quantity	Formula
Area of a circle	$A = \pi r^2$
Circumference	$C = 2\pi r$
Area of a rectangle	$A = lw$
Area of a triangle	$A = \frac{1}{2}bh$
Pythagorean theorem	$a^2 + b^2 = c^2$
Special right triangles	30-60-90 $\rightarrow x, x\sqrt{3}, 2x$ 45-45-90 $\rightarrow x, x, x\sqrt{2}$
Volume: box / cylinder	$V = lwh$ $V = \pi r^2 h$
Volume: sphere / cone	$V = \frac{4}{3}\pi r^3$ $V = \frac{1}{3}\pi r^2 h$
Volume: pyramid	$V = \frac{1}{3}lwh$
Arc / circle facts	$360^\circ \cdot \frac{\text{arc length}}{C}$ 2π radians 180° in a triangle

NOT Given — Memorize These (Algebra & Functions)

Concept	Formula
Slope	$m = (y_2 - y_1) / (x_2 - x_1)$
Slope-intercept	$y = mx + b$
Point-slope	$y - y_1 = m(x - x_1)$
Quadratic formula	$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$
Discriminant	$b^2 - 4ac$ (>0 : two real, $=0$: one, <0 : none)
Vertex form	$y = a(x - h)^2 + k \rightarrow$ vertex (h, k)
Equation of a circle	$(x - h)^2 + (y - k)^2 = r^2$

Exponent rules	$x^a \cdot x^b = x^{a+b}$ $(x^a)^b = x^{ab}$ $x^{-a} = 1/x^a$
Percent change	$(\text{new} - \text{old}) / \text{old} \times 100\%$
Distance	$d = \sqrt{(x_2 - x_1)^2 + (y_2 - y_1)^2}$
Midpoint	$((x_1 + x_2)/2, (y_1 + y_2)/2)$
Avg / mean	sum of terms / number of terms
Trig (SOH-CAH-TOA)	$\sin = \text{opp/hyp}$ $\cos = \text{adj/hyp}$ $\tan = \text{opp/adj}$

Desmos Power Moves

If the question asks...	Do this in Desmos
Solve a system of equations	Type both equations, click the intersection point.
Find a vertex / max / min	Graph it, click the peak or valley.
Find roots / x-intercepts	Graph it, click where it crosses the x-axis.
Equivalent expressions	Graph original + each choice; the match overlaps.
Solve any equation	Set $f(x) = g(x)$, graph both, read the intersection.